Priyanka Tiwari

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PROFESSIONAL SUMMARY

Fourth Year ECE student specializing in compiler design, embedded ML, and low-level systems programming. Designed and shipped ICELang, a production-grade DSL with a full lexer/parser/AST pipeline and KiCad export backend. Built GXU-1, a custom GPU co-processor on the ESP32-S3 with a handbuilt binary ISA and dual-core FreeRTOS architecture. Shipped provtrack to PyPI with 94% test coverage; contributed security-critical PRs to production open-source projects, one selected for the XERJ Corpus Hub.

CORE COMPETENCIES

Languages: C/C++, Python, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS
ML / AI: TensorFlow Lite Micro, on-device inference, model quantization (INT8), edge AI, split inference, NumPy, pandas
Compiler / Systems: DSL design, AST construction, lexer/parser pipelines, PLY (Lex-Yacc), interpreter design
Embedded & Hardware: ESP32/ESP32-S3, FreeRTOS, R-2R DAC, VGA signal generation, KiCad, Ngspice, TinyML
Frameworks & Tools: React, Vite, Next.js, Flask, FastAPI, Supabase, Matplotlib, Docker, WebSocket, REST APIs
Developer Tools: Git, Linux, VS Code, KiCad Scripting API, WebGL

PROFESSIONAL EXPERIENCE

FOSSEE Summer Intern | May 2026 – July 2026
IIT Bombay – Free and Open Source Software for Education *Remote*

- Designed and implemented ICELang, a DSL for hardware description, with a complete lexer/parser pipeline and AST, reducing manual schematic authoring time for structured hardware definitions.
- Accomplished end-to-end KiCad schematic export by translating ICELang AST nodes into native `.kicad_sch` format, automating symbol placement, wire routing, and pin mapping at ∼90% layout accuracy.
- Developed an SVG symbol rendering engine for pre-export visual verification of schematic components, eliminating repeated KiCad round-trips and measurably reducing debug cycle time.
- Integrated 50+ standard component symbols and authored test suites achieving >85% coverage, surfacing 12+ edge-case bugs before milestone submission.
- **Stack:** Python, PLY (Python Lex-Yacc), KiCad Scripting API, SVG rendering, AST design patterns, Git

PROJECTS

provtrack: Zero-Instrumentation ML Pipeline Provenance Tracker | 2026
Python, NetworkX, SHA-256, pandas, scikit-learn, PyPI

- Built a 4-layer runtime proxy (proxy → hasher → logger → graph) that instruments an entire ML session with one `activate()` call – zero decorators, zero config files, zero pipeline rewrites required.
- Engineered SHA-256 fingerprinting of every intermediate DataFrame state into a cryptographically-linked DAG; reservoir sampling caps hash latency at ∼25ms on 1M+ row inputs with deterministic cross-run hashes.
- Achieved 94% test coverage across 64 tests and 3 suites, documenting 15 edge cases (pickle bypass, inplace mutation, multi-threaded logger safety); shipped `report / export / query / diff / validate` CLI.
- Published to PyPI (`pip install provtrack`) under Apache 2.0; immutable frozen dataclass records and fully reversible `deactivate()` for production-safe deployment, zero use of `eval`/`exec`.

GXU-1: Custom GPU Co-Processor | 2026
ESP32-S3, FreeRTOS, C, React, Vite, WebSocket

- Architected a custom GPU co-processor on the ESP32-S3 with a handbuilt R-2R resistor ladder DAC for analog VGA signal output, achieving functional display output under a Rs.700 bill of materials.
- Designed a custom 8-opcode binary ISA (DRAW, FILL, CLEAR, BLIT, RECT, LINE, PIXEL, NOP) interpreted by a software rasterizer, achieving ∼30 FPS via dual-core FreeRTOS (Core 1: render, Core 0: host comms).
- Built a React/Vite WebSocket dashboard with live VRAM heatmap telemetry, opcode throughput graphs, and a browser-based ISA emulator enabling hardware-free development.

OPEN SOURCE CONTRIBUTIONS

aayushch/laya – AI Notification Command Center (Tauri + Python) | 3 merged PRs, 2026

- PR #58 – Eliminated credential leakage in error logs: stopped the engine from echoing raw request bodies (API keys, connection strings) on Pydantic validation failures; 6 files, ∼460 lines. **Selected for the XERJ Corpus Hub** as a reference implementation for AI coding agents.
- PR #40 – Closed unauthenticated loopback access to the engine REST API via OS keychain-backed bearer token auth (3 new security modules); all endpoints return `HTTP 401` on invalid tokens.
- PR #65 – Hardened filesystem permissions on sensitive application data directories to block unauthorized local reads.

ra3xdh/qucs-s – Open-Source Circuit Simulator | PR #1728 2025
Added legend support for 2D and 3D Cartesian diagram types, a missing feature in the simulation plotting engine.

ACHIEVEMENTS

Microsoft Codess Mentee 2025 – Selected for Microsoft's mentorship program for women in technology; chosen for technical merit and leadership potential.
HPAIR 2026 Delegate – Selected as a delegate for the Harvard Project for Asian and International Relations; placed in the top 1% of the global applicant pool.

EDUCATION

Acharya Institute of Technology Bangalore, IN
Bachelor of Engineering in Electronics and Communications Engineering Aug 2023 – May 2027